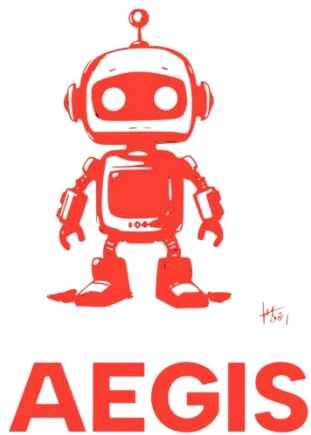


Non Functional Requirements



NFR1.1.1 Load and Concurrency

The system must support at least **200** simultaneous active users for **10** minutes without crashing, while maintaining an error rate below **1%**.

NFR1.1.2 Minimal Lag

For core user actions such as login, viewing assessments, and submitting answers, **95%** of requests must complete within **800ms** under **200** concurrent users.

NFR1.2.1 Scalability

When user load increases from **50** to **200** concurrent users, the system must scale within **5 minutes** and keep **p95** response time within **20%** of the baseline.

NFR2.1.1 Error Handling

100% of user facing validation, authorisation, and server errors must return a clear message that states what went wrong and what the user should do next and must not expose stack traces.

NFR2.2.1 Accessibility

The user interface must meet **WCAG 2.1 AA** requirements, including full keyboard navigation, contrast ratio of at least **4.5:1** for normal text, and accessible labels for all form controls.

NFR3.1.1 Access Control

100% of protected endpoints must enforce role based access control so that candidates can access only assessments assigned to them, and only recruiters can access admin functions. Unauthorized access attempts must return **403** or **404**.

NFR4.1.1 Recovery

After a critical crash, the system must be restored to service within **5 minutes**, with no loss of confirmed submitted data and no more than **30 seconds** of in progress data loss.

NFR4.2.2 MTTR

The average Mean Time to Restore for critical component failures must be **15 minutes** or less, measured over a monthly incident review period.

NFR5.1.1 Modularity

The codebase must have **zero circular dependencies** between major modules, and new features should be implementable by updating separate layers such as API, service, and data access without mixing responsibilities in one file.

NFR5.2.1 Code Quality

The project must pass linting with **zero errors**, type checking with **zero errors**, and maintain at least **80%** unit test coverage on core business logic.